ORIGINAL RESEARCH ARTICLE

High-Temperature Solutionization and Interfacial Stability of Alloy 690/Carbon Steel Clad Plates

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This work explores the influence of solutionization temperature on microstructural features at the interface of vacuum hot roll-bonded Alloy 690/ASTM SA516 GR60 carbon steel cladding. The initial as-rolled state reveals strong bonding, but also a carbide-enriched zone within the nickel alloy—specifically Cr23C6—posing a detriment to corrosion resistance. A combined approach involving diffusion modeling and thermodynamic analysis was used to optimize the solutionization regime. While theory suggested higher temperatures would mitigate carbide precipitation, experimental findings showed that a 1373 K, 4-hour treatment actually increased the width of the carbide-rich zone due to enhanced mobility of chromium and carbon. Only at 1473 K for four hours did the extent of Cr-carbides appreciably diminish. Additional Ni diffusion at elevated temperatures resulted in a distinct c-FCC interlayer on the carbon steel side, which was accentuated at 1473 K. These observations underscore that traditional solutionization temperatures, adequate for monolithic Alloy 690, require adjustment when applied to clad architectures to effectively suppress undesirable carbide precipitation.

I. INTRODUCTION

The higher SCC resistance of Alloy 690 as compared to Alloy 600 has been attributed to its relatively higher Cr content.[4] It also finds numerous applications in various components of nuclear reprocessing plants such as Joule melters for glass vitrification of high-level nuclear waste (HLW) due to its higher corrosion resistance to molten glass at high temperatures.[2] Higher corrosion resistance in acidic environments makes this a suitable material of construction for many equipment in aggressive acid production units.[3] However, the presence of costly alloying elements such as Ni and Cr in high quantities makes Alloy 690 much costlier as compared to steels and limits its widespread use in various industrial applications as a structural material. Therefore, a partial replacement of Alloy 690 in combination with a cheaper grade of steels, such as carbon steels, can economize the use of this material while exploiting its useful properties. An excellent, industrially scalable, high throughput and cost-effective method to achieve this partial replacement is to clad Alloy 690 over carbon steel through hot roll bonding (HRB). HRB is usually carried out under a protective atmosphere such as Argon,[5] Hydrogen,[6] or vacuum[7–9] to prevent oxide formation at the joining interface and achieve a sound metallurgical bonding. Among these, the vacuum hot roll bonding (VHRB) process developed at JFE corporation, Japan[10] is the most prevalent industrial practice now a days, where a package assembly consisting of cladding and backing materials is hot rolled while maintaining a static vacuum at the contact surfaces. While reports on the cladding of Ni-based superalloys through electron-beam powder bed fusion,[11] laser melting,[12] hot pressing,[13] and weld overlay[14] are widely available in literature, report on VHRB of Ni-superalloys remains scarce[8] and not at all available for Alloy 690. Moreover, cladding of Alloy 690 using techniques that involve melt deposition of Alloy 690 over a substrate results in coarse dendritic microstructure[11,15] of the Alloy 690, which may lead to poor mechanical properties of the clad layer. A major concern while using these cladding techniques is the significant dilution of composition at the interface due to intermixing of substrate and Alloy 690 leading to hot cracking at the interface.[14] Moreover, a wide heat-affected zone also forms on the substrate side,[14] which significantly deteriorates the mechanical properties of the cladding as well as the backing layers. Therefore, these cladding techniques can only be used for providing the corrosion resistance but the deposited layer cannot be used as a load-bearing structural member. Recently, heterogeneous materials[16] having a good combination of ductile and brittle material is being investigated to circumvent the inverse relation between strength and ductility. In this regard, VHRB being essentially a solid-state joining process having a narrower width interface can provide an excellent avenue to impart corrosion resistance to carbon steels through cladding Alloy 690 while improving the mechanical properties along with significant cost savings.

However, one potential challenge in VHRB of Alloy 690 and carbon steel can be the formation of a Cr-rich precipitation zone near the interface due to the diffusion of carbon from the carbon steel side toward the Alloy 690, which can deteriorate the corrosion as well as mechanical properties of the material specially at the interface. This

phenomenon has been observed to occur in the Inconel 625 and Hastelloy by Lopez et al.[13] and Tachibana et al.,[17] which ultimately affect the corrosion resistance at the interface. Thus, post rolling solutionization treatment becomes absolute necessity to minimize the impact of these carbides. Industrially, bulk Alloy 690 is mill annealed at 1373 K (1100 C) to dissolve these carbide precipitates, however, this temperature may not be sufficient enough to dissolve the Cr-rich carbides due to their increased stability with increasing carbon concentration. Therefore, identification of an optimized solutionization heat treatment either through experimentation or in combination with theoretical calculation is of absolute necessity to minimize the impact of carbide precipitation on the final product.

Therefore, in the present study an attempt has been made to fabricate a steel clad plate having Alloy 690 as a cladding layer and ASTM A516 GR60 (hereafter will be referred to as GR60) as a backing layer through VHRB technique. GR60 was chosen as the backing layer as it is widely used as materials of construction for manufacturing of pressure vessels in numerous industrial applications.[18] Thermodynamic calculations were utilized to identify suitable temperature regime for post rolling solutionization treatment in order to minimize the width of the precipitation zone. Simulation of multicomponent diffusion process during hot roll bonding and solutionization heat treatment was carried out to elucidate the microstructural evolution at the interface.

II. MATERIALS AND METHODS

A. Starting Materials

Alloy 690 and GR60 were utilized as the cladding and backing materials, respectively, for the fabrication of the clad plate. The nominal compositions (in wt pct) of both these starting materials are listed in Table I. Both these materials possess widely different microstructures, phase transformation characteristics, and mechanical and thermophysical properties. Alloy 690 shows a stable FCC-austenitic microstructure [Figure 1(a)] throughout the temperature range from RT to melting point along with minor carbide precipitates. On the other hand, GR60 shows a BCC-ferritic (a)-pearlitic microstructure at RT [Figure 1(b)], which transforms into the FCC-austenitic phase (c) at a temperature above 1123 K (850 C) (see electronic Supplementary Figure S1). As a result of their distinct composition and microstructure, both these materials have widely different mechanical and thermophysical properties, which have also been summarized in Table II. Alloy 690 generally possesses a higher ultimate tensile strength (UTS), yield strength (YS), and elongation as compared to GR60. It also shows a slightly higher thermal expansion coefficient and much lower thermal conductivity as compared to the GR60.

B. Fabrication of Clad Plate and Solutionization Treatments

A triple-layered Alloy 690/GR60 clad plate was fabricated through VHRB using a sandwiched assembly (60 9 60 9 32 mm) consisting of a GR60 plate (60 9 60 9 16 mm) placed between two Alloy 690 plates (60 9 60 9 8 mm). The contact surfaces of these plates were ground using SiC emery paper and ultrasonically cleaned using acetone to obtain oxide free surfaces. The sandwiched assembly was welded from all sides using tungsten inert gas (TIG) welding and a static vacuum of ~ 10^2 torr was created using a rotary vacuum pump to minimize the chances of oxidation at contacting surfaces during VHRB. The evacuated assembly was hot roll bonded at a temperature of 1323 K (1050 C) and strain rates of 1 to 5 s^1. The hot rolling temperature and strain rate regime were chosen based on the dynamic recrystallization domain reported for Alloy 690.[20] A two high laboratory rolling mill having 300 mm diameter rolls revolving at 45 revolution per minute was used for roll bonding. Roll bonding was carried out in multiple passes until the plate thickness was reduced to ~ 8 mm (~ 75 pct reduction). The per pass thickness reduction was varied in such a way so that the strain rate can be maintained within 1 to 5 s^1. Strain rate calculations for this purpose were performed using standard formula for rolling under sticking

Table I. Nominal Compositions of Clad Layer (Alloy 690) and Backing Layer (GR 60)

Composition (wt pct)	C	Mn	Cr	Si	Ni	Mo	Ti	Fe
Alloy 690	0.03	0.5	30	0.3	balance + other trace elements	—	0.18	10.5
GR 60	0.14	0.98	<0.015	0.23	<0.015	<0.015	—	balance + other trace elements

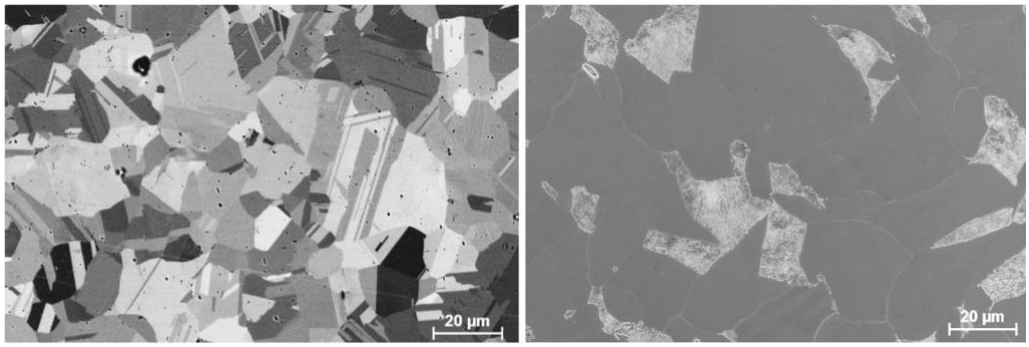

Fig. 1—Microstructures of (a) Alloy 690 and (b) GR60 starting material plates as viewed under scanning electron microscope.

Table II. Mechanical and Thermophysical Properties of Backing and Cladding Materials

Property	GR 60[18]	Alloy 690[1, 19]
Ultimate Tensile Strength (MPa)	415 to 550	650 to 750
Yield Strength (MPa)	220	280 to 450
Elongation—50 mm Gauge Length (Pct)	25	40 to 50
Thermal Coefficient of Expansion (lm/m-K)	12	14
Thermal Conductivity (W/m-K)	52	13.5

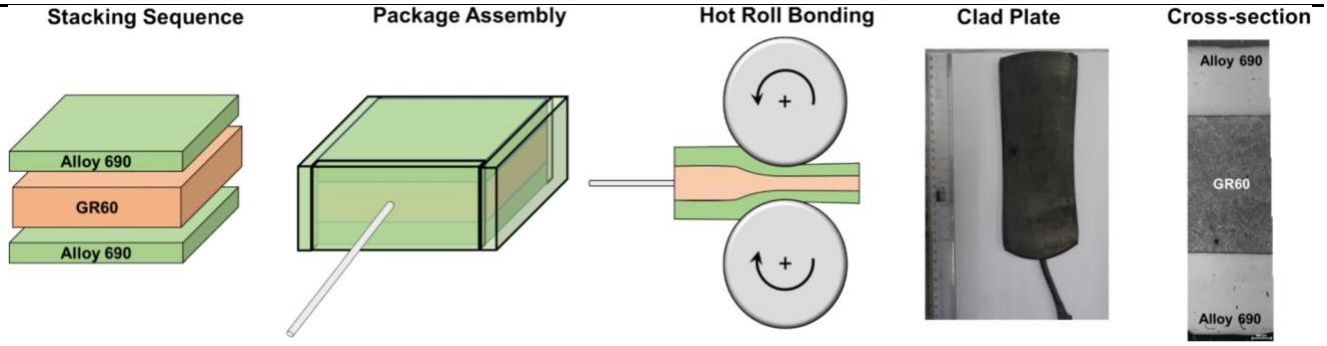

Fig. 2—Schematic illustrating the process of clad plate fabrication through VHRB technique along with the image of the fabricated clad plate and its cross-sectional microstructure.

friction condition.[21] The sandwiched assembly was soaked at 1323 K (1050 C) for 5 minutes between two subsequent rolling passes. After the final pass of rolling, the hot roll bonded clad plate was finally annealed at 1323 K (1050 C) for 15 minutes and air-cooled to homogenize the microstructure and minimize the defects associated with hot deformation. Figure 2 shows the schematic of the process for fabrication of the clad plate along with the final product. The as-rolled clad plate was subjected to two different solutionization heat treatments guided by the theoretical thermodynamic calculations to minimize the carbide precipitates. The details of these heat treatments along with their nomenclature are mentioned in Table III.

C. Characterization of Interfacial Microstructure

The microstructural evolution at the joining interface of the clad plate after post rolling heat treatments was characterized using a field emission gun source ZEISS SIGMA scanning electron microscope (FEG-SEM) equipped with backscatter electron (BSE) detector and electron backscatter diffraction (EBSD) facility. Elemental diffusion across the interface of the clad plate was quantified through compositional analysis using energy-dispersive X-

ray spectroscopy (EDS) attached to the FEG source SEM. Microstructure evolution in the GR60 layer was characterized using Carl Zeiss AxioVert A1 optical microscope (OM) and secondary electron (SE) imaging in FEG-SEM. Specimens for microstructural observations were prepared through conventional metallographic specimen preparation technique followed by a final polishing using 0.03 lm SiO_2 colloidal solution. Additionally, chemical etching with 2 pct Nital was employed for microstructural characterization of the GR60 layer. Hardness measurements were conducted using a Vickers microhardness tester using 50 g load and a 10-second dwell time to assess the effect of microstructural variation on the hardness across the joining interface.

D. Thermodynamic and Long-Range Diffusion Calculations

Thermodynamic calculations were carried out using CALPHAD methodology to find out the optimum temperature window for the solutionization heat treatments. The mc_ni.tdb, version 2.034, Ni-based alloy database and Matcalc software package[22] was utilized for these calculations. To elucidate the interfacial microstructural evolution, numerical simulation of the multicomponent long-range diffusion process during hot rolling and solutionization was carried out using the Matcalc thermokinetic software package to obtain the concentration profile across the interface.[22] The cell simulation module of the Matcalc software package, which numerically solves the one-dimensional partial differential equation governing the long-range diffusion process through the finite element method,[23] was employed to obtain the concentration profiles. The diffusion coefficients required for solving the diffusion equations were obtained from the mc_fe.ddb, version 2.015, diffusional database. The whole problem domain of 8 mm width (equal to the thickness of the clad plate) was discretized into 1600 cells, each having a width of 5 lm. These elements were further categorized into the GR60 and Alloy 690 layers by assigning the starting composition of GR60 and Alloy 690, to the left and right half of the problem domain, respectively. For simplicity, the interface between the two sections was considered as planner and only volume diffusion was considered. Simulations were carried out at different temperatures and time duration, i.e., at the rolling temperature of 1323 K (1050 C) for 1 hour and solutionization heat treatment temperatures of 1373 K (1100 C) and 1473 K (1200 C) for 4 hours. The time duration at rolling temperature was assigned 1 hour, which was the approximate time taken for the entire hot roll bonding process. In order to understand the interfacial microstructure evolution and precipitation behavior in the clad plate after different heat treatments, the composition profile obtained from the long-range diffusion calculations were then utilized to calculate the equilibrium phase fraction across the interface.

III. RESULTS

A. Microstructure

1. In As-rolled condition

Figure 3(a) shows the optical micrograph taken from the ASR specimen across the joining interface of Alloy 690 and GR60 layers. The GR60 layer revealed a graded microstructure across the thickness. Adjacent to the interface, the microstructure was mainly consisted of ferrite phase with minuscule fraction of the pearlite phase. The higher volume fraction of the ferrite phase near the interface was a direct indication of the decarburization of this region as a result of the diffusion of carbon from the GR 60 layer toward the Alloy 690. The minor amount of pearlite phase was present between ferrite grains in the form of small islands, as shown in Figure 3(b). Figure 3(c) shows the cross-sectional backscatter electron image obtain from the ASR specimen. A very few discrete oxide particles were noticed to be present at the interface between the GR60 backing layer and Alloy 690 cladding layer indicating the formation of an excellent metallurgical bonding. A closer look at the interface [as shown in Figure 3(d)] also revealed the presence of a thin layer between the decarburized ferrite region of GR60 and the carburized region of Alloy 690 having a fine lath microstructure, which was possibly a product of martensitic transformation. Similar to the GR60 backing layer, the Alloy 690 cladding layer also showed a graded microstructure due to diffusion of alloying elements closer to the interface [Figure 3(e)]. As shown in Figure 3(e), enrichment of Alloy 690 with C led to the formation of a precipitation zone (region P) having a width of approximately 100 lm. These precipitates showed a graded distribution in their volume fraction and size. The region adjacent to the interface having a width of approximately 5 to 6 lm (region PF) was devoid of the precipitates barring a few very fine precipitates of TiN, which were originally present in the starting material as well [Figure 1(a)]. Coarse precipitates began to appear at the end of the PF region. These precipitates showed a

bimodal size distribution at the beginning of the precipitation zone consisting of coarse precipitates having a size in the range of 2 to 4 lm and smaller precipitates having a size less than 1 lm. However, the volume fraction of coarser precipitates gradually decreased with increasing distance from the interface. On the other hand, the volume fraction of fine precipitates initially increased with increasing distance from the joining interface and gradually started declining toward the end of precipitation zone. Figure 4, the EDS elemental maps obtained from the precipitate zone (P), revealed that these precipitates to be rich in Cr and Mn while lean in Fe and Ni. EBSD mapping of the precipitation zone in Alloy 690 was carried out to identify the structure of these Cr-rich precipitates. EBSD phase maps [as shown in Figure 5(a)] show that

these precipitates are of $Cr_{23}C_6$ carbide having FCC crystal structure. EBSD maps also revealed that the

Table III. Heat Treatment Schedule for Fabricated Clad

Specimen Name	Heat Treatment
ASR	hot rolled at 1323 K (1050 C)
S1100	ASR + Solutionization @ 1373 K (1100 C) for 4 hours followed by water quenching
S1200	ASR + Solutionization @ 1473 K (1200 C) for 4 hours followed by water quenching

Plate coarser $Cr_{23}C_6$ precipitates mostly formed at the energetically more favorable sites such as grain boundaries or twin boundaries. In contrast, the finer precipitates were mostly confined within the grains.

EBSD maps [Figure 5(a)] also show the presence of a graded microstructure of the Alloy 690 matrix. Grains near the interface were much finer as compared to the bulk of the Alloy 690 layer. The presence of the finer grains near the interface could be attributed mainly to formation of larger number of recrystallized grains on account of a higher number of nucleation sites provided

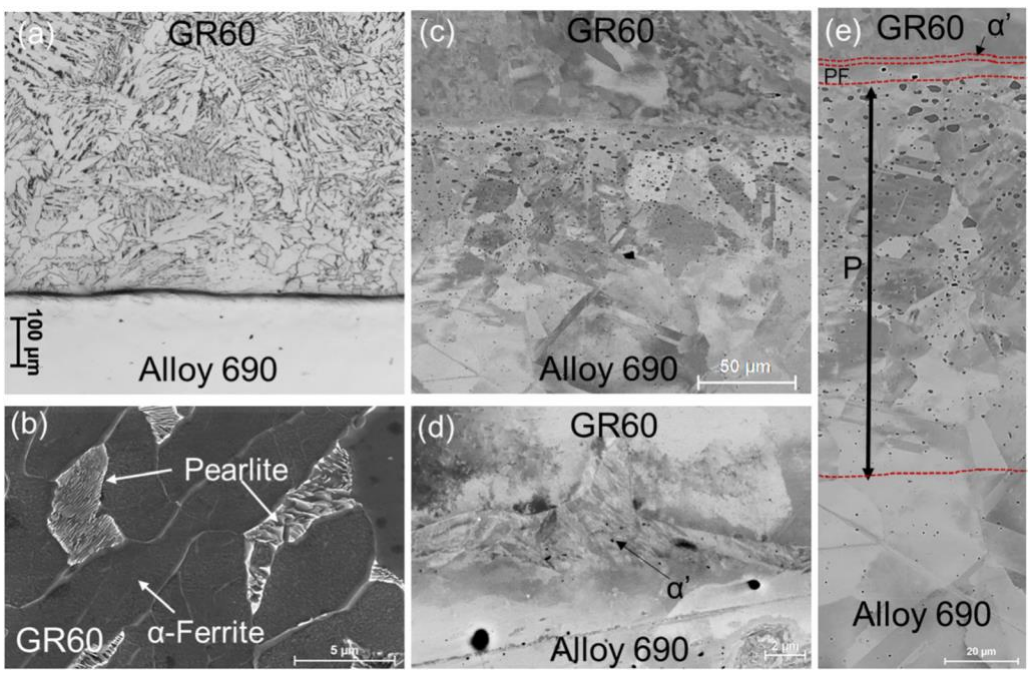

Fig. 3—(a) Optical micrograph obtained from GR60 layer of as-rolled (ASR) specimen showing a graded ferritic-pearlitic microstructure, (b) a high magnification in-lens SEM image of GR60 showing presence of pearlitic phase between ferritic grains, (c) Backscatter electron (BSE) image of interfacial region of the Alloy 690/GR60 clad plate showing a very good metallurgical bonding between Alloy 690 and GR60 with the presence of very few oxide particles, (d) a high magnification image of the Alloy 690/GR60 interface revealing presence of a thin martensitic layer (a'), and (e) BSE image of Alloy 690 layer showing a formation of precipitation zone having a graded distribution of precipitates.

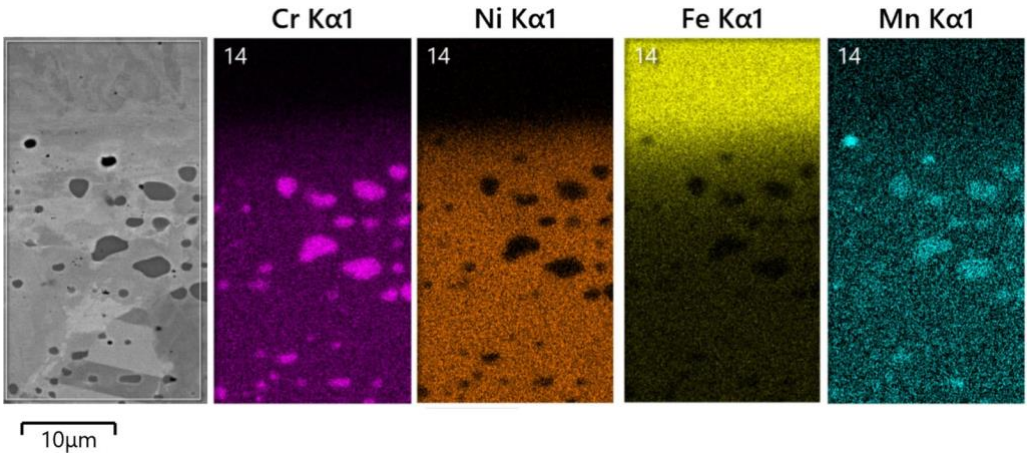

Fig. 4—SEM-EDS elemental distribution maps of the precipitation zone in Alloy 690 cladding layer of ASR specimen showing presence of precipitates rich in Cr and Mn, and lean in Ni and Fe.

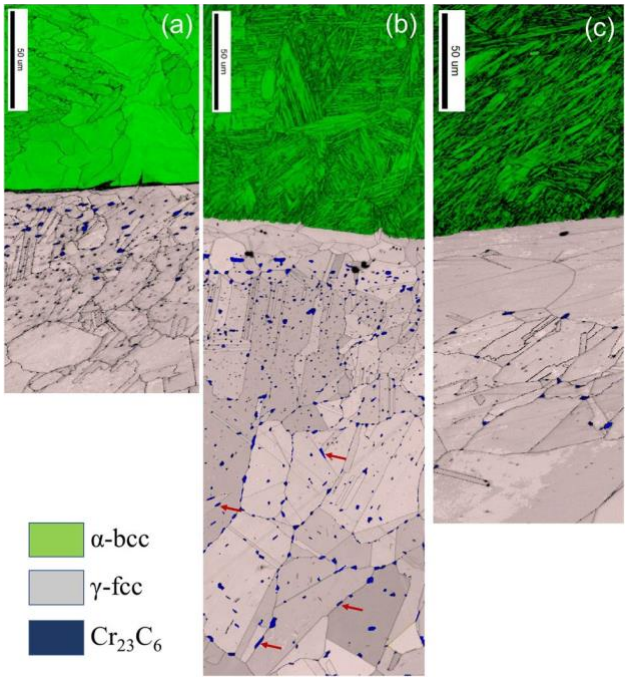

Fig. 5—EBSD phase maps obtained from (a) as-rolled (ASR), (b) solutionized at 1373 K (1100 C)-4 h (S1100), and (c) solutionized at 1473 K (1200 C)-4 h (S1200) specimens revealing the presence of $Cr_{23}C_6$ carbides on Alloy 690 side.

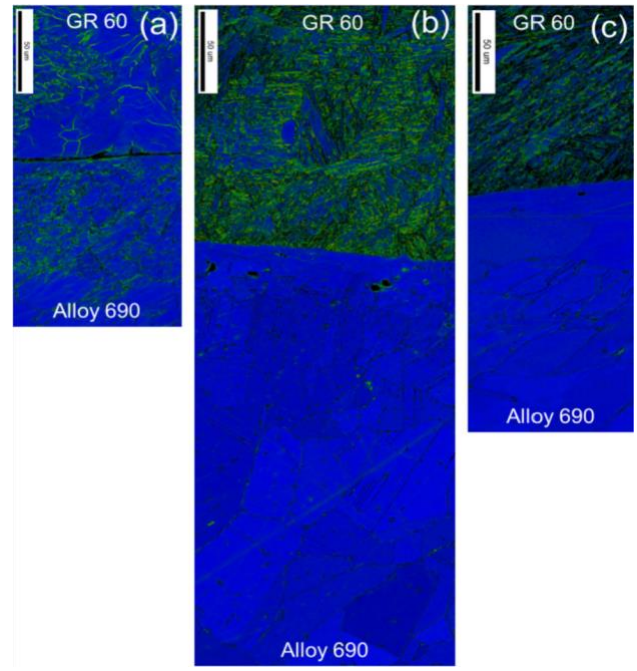

Fig. 6—EBSD kernel average misorientation maps of (a) as-rolled (ASR), (b) solutionized at 1373 K (1100 C)-4 h (S1100), and (c) solutionized at 1473 K (1200 C)-4 h (S1200) specimens showing presence of GNDs in Alloy 690 layer in ASR specimen which gets annihilated during subsequent solutionization heat treatments.

by geometrically necessary dislocations (GNDs). The generation of a significantly greater number of GNDs near the interface can be attributed to the strain incompatibility between GR60 and Alloy 690 while hot roll bonding. The presence of the residual GNDs is clearly evident in KAM map [Figure 6(a)] showing a clear contrast between interfacial and bulk regions. Moreover, the simultaneous precipitation of $Cr_{23}C_6$ particles restricted the growth of these recrystallized grains within the precipitation zone of Alloy 690.

2. After solutionization treatment

The precipitation of Cr-rich carbides present mostly on grain boundaries in Alloy 690 depletes the amount of Cr near grain boundaries and results in the degradation of intergranular corrosion (IGC) resistance. Therefore, a solutionization treatment is necessary to dissolve these carbides and restore the amount of Cr in the matrix. Industrially, Alloy 690 is annealed in range 1323 to 1373 K (1050 to 1100 C)[4,24] to reduce the amount of carbide particles. However, this temperature, as will be shown later in this work may not be adequate to dissolve these carbides in the interface region of Alloy 690 due to increased carbon content as a result of diffusion from the GR60 side. This point can be more clearly understood from the computed carbon isopleth of Alloy 690 as shown in Figure 7 showing the stability of various equilibrium phases at different temperatures and carbon concentration. As evident, at 1373 K (1100 C) the FCC phase shows a very little solubility for C (only 0.014 wt pct) and the excess carbon precipitates out as $Cr_{23}C_6$ carbide. Therefore, a significant fraction of the $Cr_{23}C_6$ carbide phase may be present even after solutionization at 1373 K (1100 C) due to the increased amount of carbon diffusion from GR60. The solubility of C increases with increase in temperature and reaches its maximum solubility at ~ 1588 K (1315 C), which is close to the liquidus temperature. Moreover, $Cr_{23}C_6$ carbide also get converted to Cr_7C_3 carbide as a stable carbide phase at high temperatures. Therefore, it is expected that solutionization at temperatures close to 1573 K (1300 C) can lead to a substantial reduction in the amount of carbide. However, solutionization at such a high temperature may lead to significant grain coarsening of both Alloy 690 and GR60 resulting in poor mechanical properties. Therefore, an optimum temperature for solutionization around 1473 K (1200 C) followed by rapid cooling could be an effective solution to reduce the amount of carbide (as complete elimination is not possible) as well as achieving optimum mechanical properties. It is to be noted that a significant area of the diffusion zone between Alloy 690 and GR60 is expected to have a Cr concentration lower than the bulk of Alloy 690 (<30 wt pct). The computed carbon isopleth at a lower Cr concentration (i.e., 15 wt pct) clearly shows that the reduction in Cr concentration can substantially increase

the solubility of carbon in FCC-c and result in a PF region (which appears as a thin layer between joining interface and precipitation region in the ASR specimen). Therefore, solutionization in the 1373 K to 1473 K (1100 C to 1200 C) temperature range can substantially increase the width of this PF region and results in a near carbide-free interfacial region.

Based on the calculated phase diagram using thermodynamic data, the as-rolled clad plate was solutionized at two different temperatures, i.e., 1373 K (1100 C) and 1473 K (1200 C) for 4 hours to dissolve the Cr-rich carbides. The selection of 1373 K (1100 C) solutionization temperature was mainly based on the annealing temperature usually employed during industrial processing of Alloy 690 and to illustrate the effect of increased carbon concentration on the carbide precipitation.[1] As discussed in the previous paragraph, the S1100 specimen solutionized at 1373 K (1100 C) showed [Figure 8(a)] a marked increase in the precipitation of Cr-carbide as compared to the ASR specimen. The width of the precipitation zone (region P) increased from ~ 100 to ~ 250 lm, while that of precipitate-free (region PF) increased from ~ 5–6 to 16–18 lm. The Cr-rich carbides in the precipitation zone were present in both grain boundaries as well as grains interior. However, grain boundary carbides were significantly coarser as compared to those present in the interior of the grains. The average size of these Cr-carbide precipitates showed a V-shaped trend, i.e., coarse precipitates mostly present at the grain boundaries near the interface followed by intragranular fine precipitates in the middle and finally coarse intergranular precipitates within the bulk of the specimen. The shape of these Cr-carbides near the interface was mostly spherical, however, a significant number of carbide precipitates with a plate shape were observed toward the bulk of Alloy 690 [Figure 8(c)]. The EBSD phase map [Figure 5(b)] elucidated that both these spherical and plate-shaped precipitates were structurally identical and identified to be fcc-$Cr_{23}C_6$ carbide phase. As evident from the ESBD map, the plate-shaped $Cr_{23}C_6$ carbides were mostly present on the twin boundaries and grain interiors, while the spheroidal-shaped carbides were present on the high-angle grain boundaries. This is in good agreement with the reported literature wherein the shape of $Cr_{23}C_6$ precipitates in Alloy 690 has been shown to depend on the misorientation angle of interface boundaries.[25] The high-angle boundaries having higher energy as compared to twin boundaries provides a preferential site for nucleation of $Cr_{23}C_6$ carbides. The $Cr_{23}C_6$ carbides nucleate on the high-angle grain boundaries maintaining a cube to cube ($(111)_M//(111)_P$) orientation relationship (OR) with one of the grains involved.[26] They grow along the grain boundary as well as into the other grain without maintaining an OR, thereby attaining a spheroidal shape.[26] The nucleation as well as growth of $Cr_{23}C_6$ precipitates at twin boundaries is suppressed due to their lower energy leading to a needle/plate shape morphology aligning along the twin boundaries. Moreover, precipitates nucleating within the grain interior have plate shape morphology due to necessity of maintaining the specific crystallographic OR with the matrix phase (i.e., cube to cube OR) to minimize the interfacial energy.[27]

A significant enhancement in the degree of precipitation indicated that the solutionization temperature of 1373 K (1100 C) was inadequate to dissolve the $Cr_{23}C_6$ precipitates. Therefore, the solutionization temperature was further raised to 1473 K (1200 C) while keeping the duration of heat treatment same. As shown in Figures 9(a) and (c), clad specimen solutionized at 1473 K (1200 C) for 4 hours (S1200) showed a much lower degree of precipitation as compared to ASR and specimens solutionized at 1373 K (1100 C) (i.e., S1100). The width of the precipitation zone (region P) in the S1200 specimen was also found to decrease from ~ 250 lm in the S1100 specimen to ~ 100 lm, which was similar to that observed in the ASR specimen. However, the precipitation zone shifted away from the interface toward the bulk of Alloy 690 and the fraction of precipitation also decreased. The width of the precipitate-free (PF) zone also increased from 16 to 18 lm in the S1100 specimen to 45 to 50 lm in the S1200 specimen. The EBSD phase map [Figure 5(c)] reveals that these precipitates were also $Cr_{23}C_6$ carbide precipitates similar to that was observed in the ASR and S1100 specimens. These precipitates had a spheroidal shape [Figure 9(c)] and were predominately present on the grain boundaries, although, a small amount of fine intragranular precipitates could still be noticed. As shown in Figure 5(c), a significant grain coarsening could also be observed in specimen S1200, which could be attributed to a higher solutionization temperature as well as to the dissolution of the $Cr_{23}C_6$ carbide particle, which acted as pinning sites for restricting the grain boundary mobility in the case of other two specimens. In both the solutionized specimens, the residual stress due strain incompatibility between Alloy 690 and GR60 was significantly lower than the as received condition as evident in the KAM maps [Figures 6(b) and (c)].

Another important observation from Figures 8(a) and 9(a) is the migration of the interface boundary toward the GR60 side. The apparent interface boundary identified based on differing microstructure on either side and the original interface boundary usually decorated with oxide particles have been delineated using red and yellow lines, respectively in Figures 8(a) and 9(a). In the case of ASR specimens, no such separation existed between the apparent and actual interfaces and these two nearly coincided with each other. However, this separation became pronounced after solutionization and a wider separation was noticed in the case of the S1200 specimen. The probable reason for this phenomenon appears to be the stabilization of the FCC phase within the GR60 layer due to enhanced Ni diffusion, which will be elaborated in further subsequent sections.

As far as the bulk microstructure of GR60 is concerned significant changes occurred after solutionization. Moreover, in contrast to the ASR specimen, the interface in the S1100 [Figure 8(e)] and S1200 [Figure 9(e)] specimens appeared to be much sharper and showed no distinct thin layer of lath martensite as observed in the case of the ASR specimen. Since both the solutionized specimens were water

quenched to RT from solutionization temperature at a relatively higher cooling rate to avoid precipitation of carbides, a martensitic microstructure was observed in the GR60 layer throughout the entire width extending till the interface as noticed in Figures 8(b) & (d) and 9(b) & (d). This is tantamount to the extension of the thin martensite layer present in the ASR specimen to the entire width of the GR60 layer after solutionization.

IV. DISCUSSION

A. Precipitation Behavior

The precipitation behavior observed in the as-rolled as well as solutionized conditions can further be elucidated with the help of the quantitative EDS analysis across the interface between Alloy 690 and GR60. As shown in Figure 10, EDS composition measurements in all the specimens revealed diffusion of Cr and Ni from Alloy 690 side toward the GR60 layer and diffusion of Fe in the opposite direction. This diffusional movement of different solute elements occurred along the concentration gradient in all specimens. Although, a noticeable difference in the diffusion distance could be observed in different types of specimens. For example, the diffusion distance of Ni was found to increase from ~ 30 lm in as-rolled specimen (ASR) to ~ 45 lm in the 1373 K (1100 C) solutionized specimen (S1100) and ~ 90 lm in the 1473 K (1200 C) solutionized specimen (S1200). This along the expected line as the mobility of solute atoms is expected to increase with increase in solutionization temperature, which resulted in a wider diffusion zone. Additionally, the diffusion distance of Cr was found to be slightly higher as compared to Ni in all the specimens but showed a similar trend. Occasionally, spikes in Cr concentration and simultaneous dip in Ni and Fe concentrations could also be observed in Figure 10 which is essentially due to the presence of Cr-rich carbide precipitates. It is to be noted that C concentration could not be measured accurately using the EDS technique and therefore, has not been included in these figures. However, a theoretical calculation utilizing the Matcalc thermos-kinetic software package was carried out to estimate the variation in carbon concentration and understand the precipitation behavior across the interface. Figures 10(a) through (d) show the numerically simulated concentration profiles of Fe, Cr, Ni, and C superimposed on top of the experimentally measured concentration profiles of Fe, Cr, and Ni. A reasonably good agreement between simulated and measured concentration profiles of Fe, Cr, and Ni could be observed. The computed C concentration profile [Figure 10(d)] shows the presence of a carbon-depleted zone near the interface on the GR60 side and a carbon-enriched zone on the Alloy 690 side. The occurrence of uphill diffusion of carbon across the interface resulting in a spike in the carbon concentration at the interface was also observed.

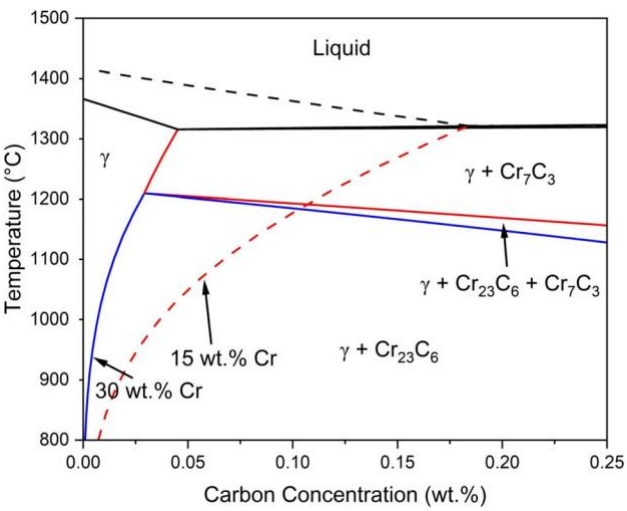

Fig. 7—Carbon isopleth of Alloy 690 showing solubility limit for carbon in c-FCC phase and associated thermodynamically stable carbide phases. Dotted lines show curves for 15 wt pct Cr.

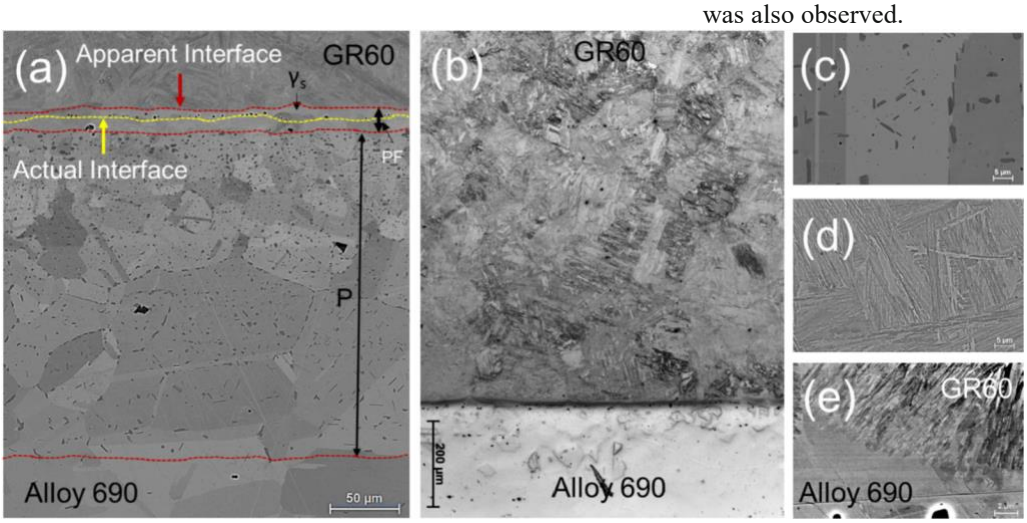

Fig. 8—(a) BSE image of the interfacial region of specimen solutionized at 1373 K (1100 C) (S1100) showing an increase in width of the precipitation zone, (b) optical micrograph of GR60 layer showing the presence of a martensitic microstructure, (c) a high magnification BSE image of precipitation zone in Alloy 690

showing presence of rod shaped Cr-carbide precipitates, (d) a high magnification image showing martensitic microstructure of GR60, and (e) BSE image of Alloy 690/GR60 interface (apparent) showing a sharp microstructural transition from fcc-c to bcc-a'.

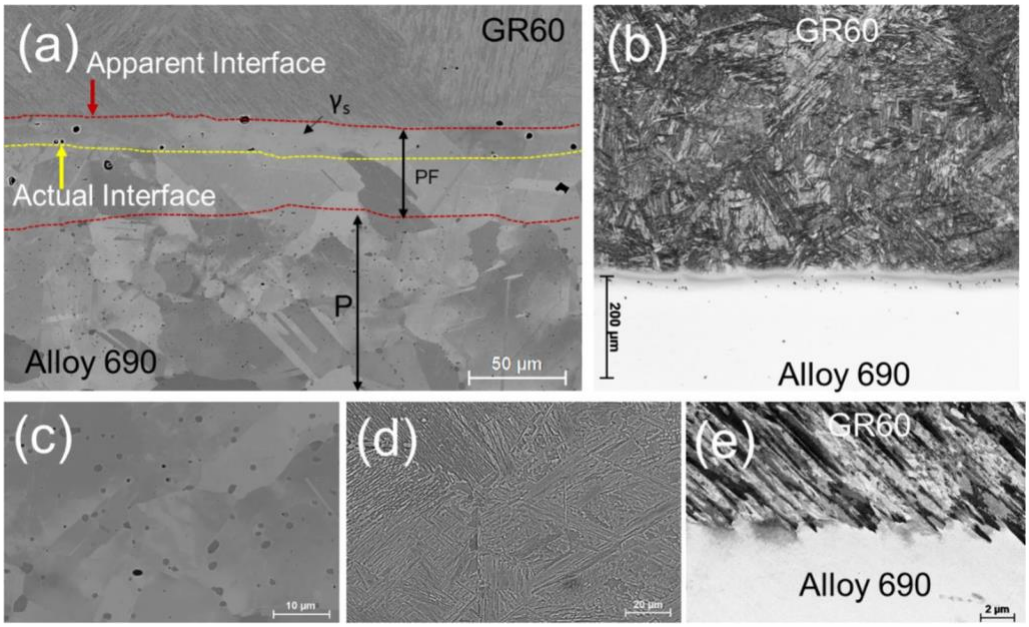

Fig. 9—(a) BSE image of interfacial region of specimen solutionized at 1473 K (1200 C) (S1200) showing a decrease in the degree of precipitation, (b) optical micrograph of GR60 layer showing the presence of a martensitic microstructure, (c) a high magnification BSE image of precipitation zone in Alloy 690 showing presence of spheroidal-shaped $Cr_{23}C_6$ precipitates, (d) a high magnification image showing the martensitic microstructure of GR60, and (e) BSE image of Alloy 690/GR60 interface showing a sharp microstructural transition from fcc-c to bcc-a'.

In order to understand the precipitation behavior across the joining interface, the equilibrium phases at different points were calculated using the CALPHAD methodology utilizing these simulated composition profiles. Figure 11 shows the variation of equilibrium phase fraction across the joining interface in different clad specimens. In the ASR specimen, thermodynamic calculations predicted the presence of the c-fcc phase as a major phase throughout the cross-section of the clad specimen at 1323 K (1050 C). Similarly, the presence of $Cr_{23}C_6$ as the second phase was also predicted in the Alloy 690 side whose volume fraction rises rapidly near the interface. The mole fraction of $Cr_{23}C_6$ began to decrease after attaining a peak near the interface and became very small in the bulk region of Alloy 690. This inverted 'V' shaped distribution of $Cr_{23}C_6$ predicted by thermodynamic calculations was broadly in line with the experimental observations made in the ASR specimen [Figure 3(e)] and occurred due to an inverse relation between the concentration profiles of Cr and C (Figure 10). Thermodynamic calculations also predicted a similar precipitation behavior at 1373 K (1100 C), although, the mole fraction of $Cr_{23}C_6$ was found to be lower than that of ASR specimen. However, it also showed a wider carbide precipitation zone having $Cr_{23}C_6$ precipitates due to increased carbon diffusion distance, which explains the increase in the width of the precipitation zone experimentally observed in specimen S1100 [Figure 8(a)]. Additionally, a small zone having Cr_7C_3 precipitates near the interface was also predicted by thermodynamic calculations instead of a PF zone observed experimentally. On the other hand, thermodynamic calculations predicted a complete absence of the $Cr_{23}C_6$ phase with the presence of a minor fraction of the Cr_7C_3 phase in specimen solutionized at 1473 K (1200 C) S1200. Additionally, thermodynamic calculations also predicted a precipitate-free zone of ~ 25 lm near the interface at 1473 K (1200 C) which matches well the experimental observation, although, the width was significantly smaller as compared to observed 45 to 50 lm wide PF zone.

The preceding discussion clearly showed that the stability of $Cr_{23}C_6$ decreases with a decrease in Cr concentration and increase in heat treatment temperature. The lower stability of $Cr_{23}C_6$ with a decrease in Cr concentration can be deduced from Figure 12(a) which shows the thermodynamically calculated phase fraction of $Cr_{23}C_6$ and Cr_7C_3 with varying Cr concentration. This can also be realized intuitively as the Cr concentration in $Cr_{23}C_6$ is almost 1.6 times that in Cr_7C_3 thus lower Cr concentration will definitely favor the formation of Cr_7C_3 phase. The lower stability of $Cr_{23}C_6$ at high temperatures was also brought out by the carbon isopleth of Alloy 690 (Figure 7) which showed that $Cr_{23}C_6$ gets replaced by Cr_7C_3 at high temperatures. However, the presence of Cr_7C_3 precipitates could not be experimentally substantiated in as-rolled as well as in solutionized specimens possibly on account of slower precipitation kinetics of Cr_7C_3 phase and increased Fe concentration from the GR60 side. The reduction in amount of dissolved carbon in c-fcc matrix due to the precipitation of $Cr_{23}C_6$ phase while rolling may also have led to a situation where the driving force was insufficient to overcome the

energy barrier for nucleation of Cr_7C_3 phase. Thermodynamic calculations [Figure 12(b)] further showed that an increase in Fe

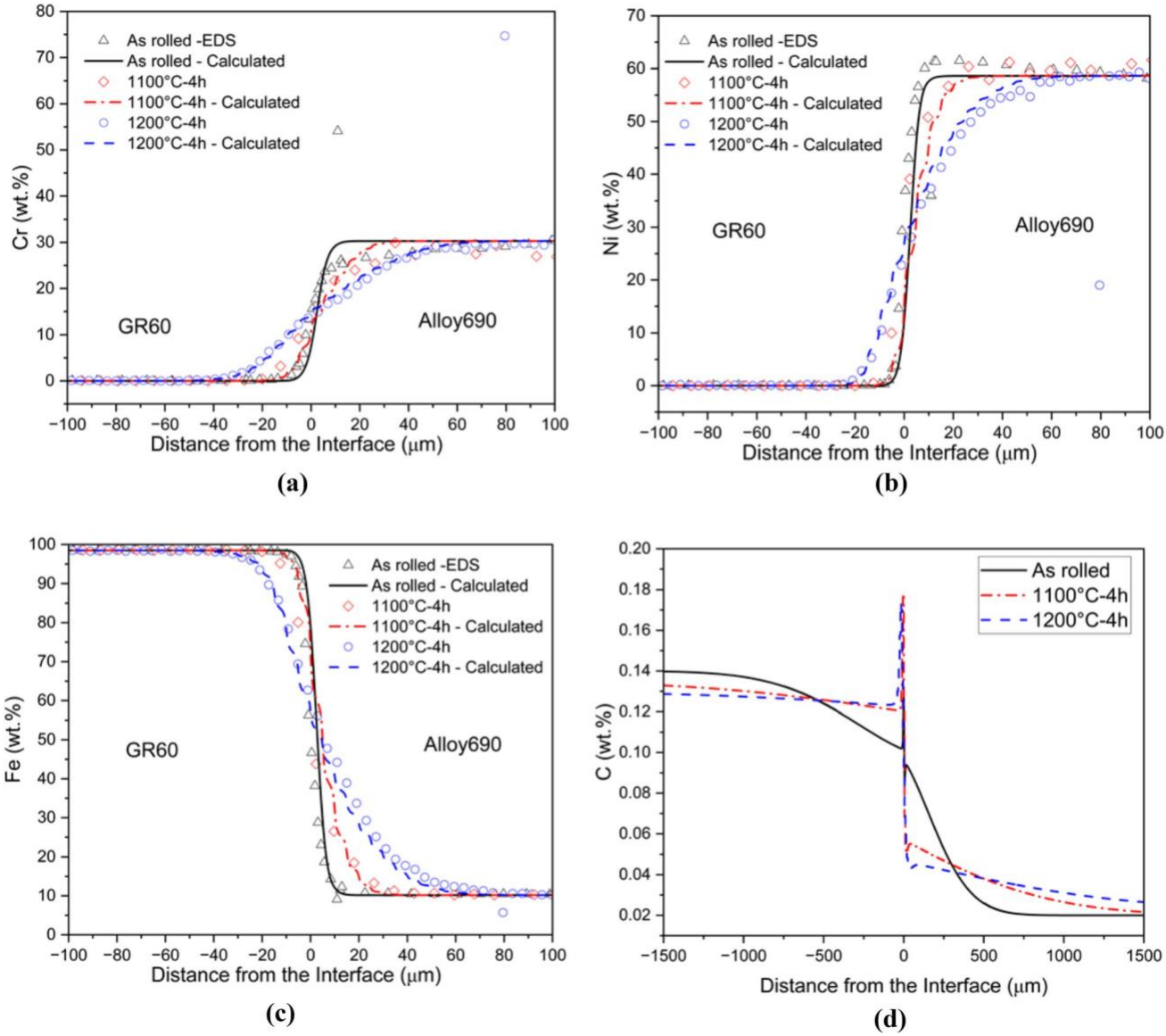

Fig. 10—Composition profiles of (a) Cr, (b) Ni, (c) Fe elements measured using SEM-EDS technique. Theoretically calculated composition profiles were superimposed on top it and (d) theoretically calculated profile of Carbon across the joining interface of Alloy 690 and GR60 in various clad plate specimens.

concentration tends to destabilize the Cr_7C_3 carbide and may as well slow down its precipitation kinetics. Thus, it looks like the volume fraction of $Cr_{23}C_6$ carbide precipitates decreased substantially after solutionization at 1200 C due to increased solubility of carbon in the c-fcc phase. However, beyond a certain extent $Cr_{23}C_6$ formed during hot roll bonding remained stable due to the slower kinetics of Cr_7C_3 carbide precipitation.

B. Evolution of Matrix Microstructure

The microstructure at the joining interface between Alloy 690 and GR60 distinctly differs in as-rolled and solutionized conditions. A thin lath martensitic layer at the interface observed in as-rolled specimen (ASR) could not be observed after solutionization and interface between Alloy 690 and GR60 becomes sharper. These microstructural differences could be attributed mainly to the difference in the extent of alloying elements diffusional movement across the interface as well as the difference in the post-processing cooling rate employed. A brief summary of the interfacial microstructural evolution along with formation of various microstructural zones across the interface has been schematically depicted in Figure 13.

As shown in Figure 13(a), the RT microstructure of GR60 and Alloy 690 starting material is composed of a-bcc and c-

fcc, respectively. While heating from RT to hot rolling temperature [1323 K (1050 C)], the GR60 layer undergoes a to c phase transformation, while Alloy 690 remains in c phase. As depicted in Figure 13(b), during hot rolling and inter-pass annealing, the Ni and Cr started diffusing out from Alloy 690 toward the GR60 layer, while Fe, C, and Mn in the opposite direction. This led to the formation of graded compositional zones across the interface as depicted in Fig. 13(c). After the final rolling pass, the bulk of GR60 layer of the ASR specimen will transform from its high temperature austenite phase to ferritic-pearlitic phases while air cooling (AC). Although, the carbon depletion (C region) within the GR60 layer near the interface will led to the formation of a ferritic microstructure. Simultaneous enrichment in Cr and Ni (Ni^+ region) at the interface due to the diffusional movement from Alloy 690 layer will increase the hardenability of this region. This increased hardenability will possibly be led to the c-fcc to martensite (a') transformation at the interfacial region while air cooling, as depicted in Figure 13(d). The thickness of this martensite layer (2 to 3 lm) was very narrow [as depicted in Figure 3(d)] due to the very limited Cr and Ni enriched region (5 lm wide) within GR60 at 1323 K (1050 C) while hot roll bonding. However, there were some locations along the interface where the thickness of this martensitic layer was relatively higher (~ 5 to 6 lm) as compared to the surrounding region due to enhanced diffusion of Cr and Ni into the GR60 layer through grain boundary diffusion as depicted in Figure 14.

During solutionization heat treatments as well, GR60 layer of ASR specimen underwent a to c phase transformation as shown in Figures 13(e) and (h). The diffusion of solute elements again led to the formation of distinct compositional zones similar to ASR specimen, although, with different widths [Figures 13(f) and (i)]. As these clad plate specimens were water quenched (WQ) after 4 hours holding at solutionization temperatures,

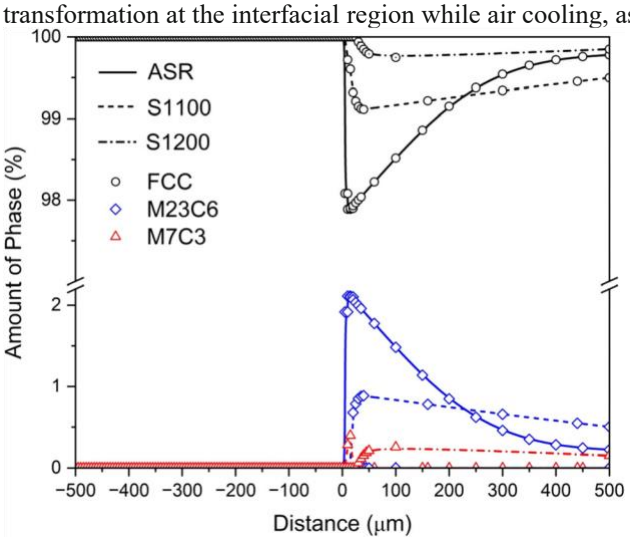

Fig. 11—Variation of equilibrium phase fraction across the joining interface of clad steel pate in different heat-treated conditions calculated through Matcalc.

the formation of martensite phase [Figures 13(g) and (j)] throughout the GR60 layer is obvious due to higher rate of cooling. Thus, the original graded ferritic-pearlitic microstructure present of the GR60 layer in ASR specimen got completely transformed to martensite. However, it could be noticed from Figures 8(a) and 9(a) as well as Figures 13(g) and (j) that the original interface, decorated by the oxide particles, did not coincide with the apparent interface (showing distinct matrix microstructure on either side) in both the solutionized specimens. The region between the apparent and original interface identified to be c-fcc (hereafter referred to as stabilized c_s) phase as depicted in Figures 13(g) and (j) and also observed in Figures 5(b) and (c). The presence of c_s between the original and apparent interface could be attributed to the enrichment of this region (originally part of the GR60 layer) in Ni (Ni^+ region) and simultaneously depletion in Fe, which may have led to the stabilization of the c_s phase even after fast cooling from solutionization temperatures. Further, the width of this c_s region increased with an increase in solutionization temperature (1 to 6 lm in the S1100 specimen to ~ 9 to 15 lm in S1200 specimen), which was a direct manifestation of the increased width of Ni enriched zone (Ni^+ region) within GR60 from ~ 15 lm in S1100 specimen to ~ 27 lm in S1200 specimen as shown in Figure 13(b). The formation of c_s phase can be justified well with the help of the Schaeffler diagram,[28] which shows the map of stable phases at RT at

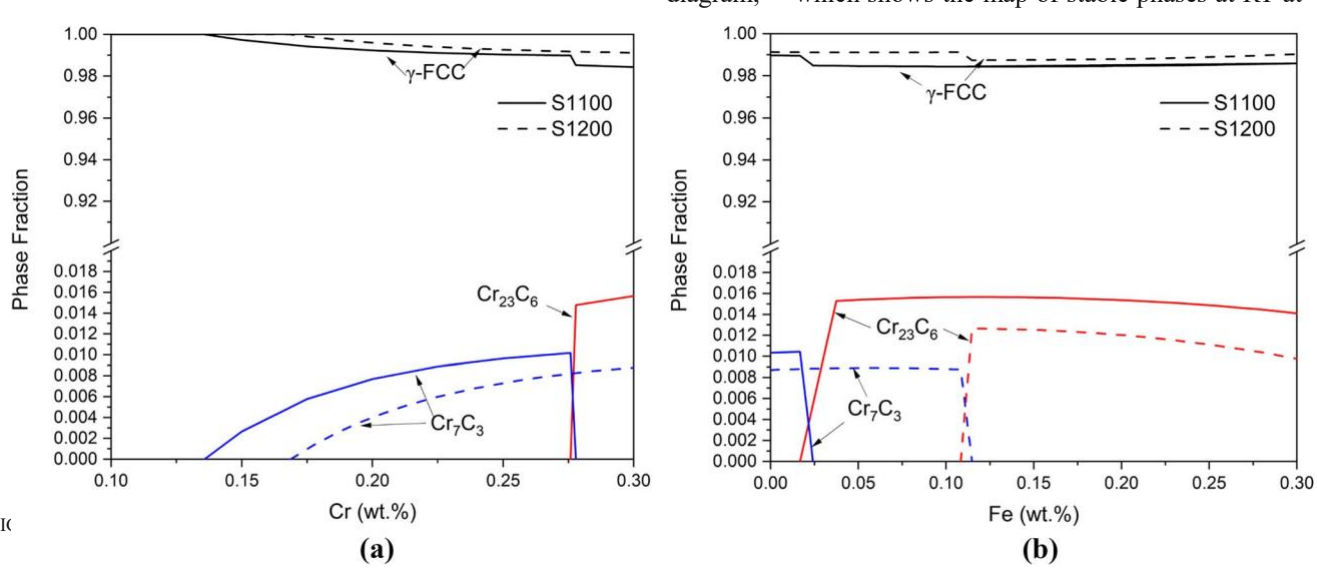

Fig. 12—Variation of equilibrium phases in Alloy 690 at different temperatures with (a) Cr and (b) Fe.

different amount of Ni_{eq} and Cr_{eq} [formulas given below respective axes in Figure 15(a)]. The Schaeffler diagram is a constitution diagram usually derived based on experimental observations and are very useful in predicting the metastable microstructure of stainless steels. To ascertain the formation of c_s, variation of Cr_{eq} for S1100 and S1200 specimens across the interface was calculated using local compositions of Cr, Si and Ti [in red dashed and solid lines, respectively in Figure 15(b)]. The minimum $Ni_{eq,min}$ required to stabilize the austenite (c-fcc) was then determined from Schaeffler diagram show in Figure 15(a) and plotted for different points across the interface. The regions having a Ni_{eq}, calculated using local C, Mn and Ni concentration in S1100 and S1200 specimens [in blue dashed and solid lines, respectively in Figure 15(b)], greater than the $Ni_{eq,min}$ are then expected to have a c_s phase. The width of such regions determined from Figure 15(b) are 2.5 and 9.5 lm for S1100 and S1200 specimen, respectively, which are within the range of experimentally measured values of width c_s region. The wide variation in experimentally measured width of c_s region can be due to the non-planar interface as well as diffusion through grain boundaries. It is to be noted that the stabilization of austenite (c_s) observed here is similar to that observed in the austenitic stainless steels such as SS304, wherein a metastable c.FCC microstructure exists at room temperature due to presence of a high amount of Ni.[29]

C. Hardness Variation Across the Interface

Microstructural variation across the Alloy 690-GR60 interface and its subsequent changes after solutionization affect the mechanical properties locally, which can be effectively deduced from the measurement of hardness variation across the interface. Figure 16 shows the hardness distribution across the interface in various clad steel specimens. In all three specimens, the region coinciding with the Cr-carbide precipitation zone within the Alloy 690 layer showed higher hardness values as compared to the surrounding regions. The higher hardness of this region could be attributed to the precipitation strengthening effect arising from the $Cr_{23}C_6$ precipitates, which generally have higher hardness. Apart from this, the hardness value observed in the precipitation zone was highest in the case of the ASR specimen followed by S1100 and lowest in the case of the S1200 specimen. This difference in hardness value among these three specimens appeared to be due to the difference in their average grain size as with increasing solutionization temperature the average grain size increased from 20 lm in the ASR specimen to 39 lm in the S1100 specimen and 44 lm in S1200 specimen. The ASR specimen having finer grains showed higher hardness values as compared to S1200 having coarser grains. Overall, the hardness values in the bulk of Alloy 690 were always lower than those in the precipitation zone in all three specimens. However, significant grain coarsening also led to lower hardness values in the bulk region of Alloy 690 layer of the solutionized specimens as compared to ASR specimens. Interestingly, as portrayed in Figure 16, on the Alloy 690 side all the specimens showed a sharp drop in the hardness values at the interface as compared to the adjacent precipitation zone. This trend is essentially associated with the carbide precipitate-free region ahead of the precipitation zone in the Alloy 690 side.

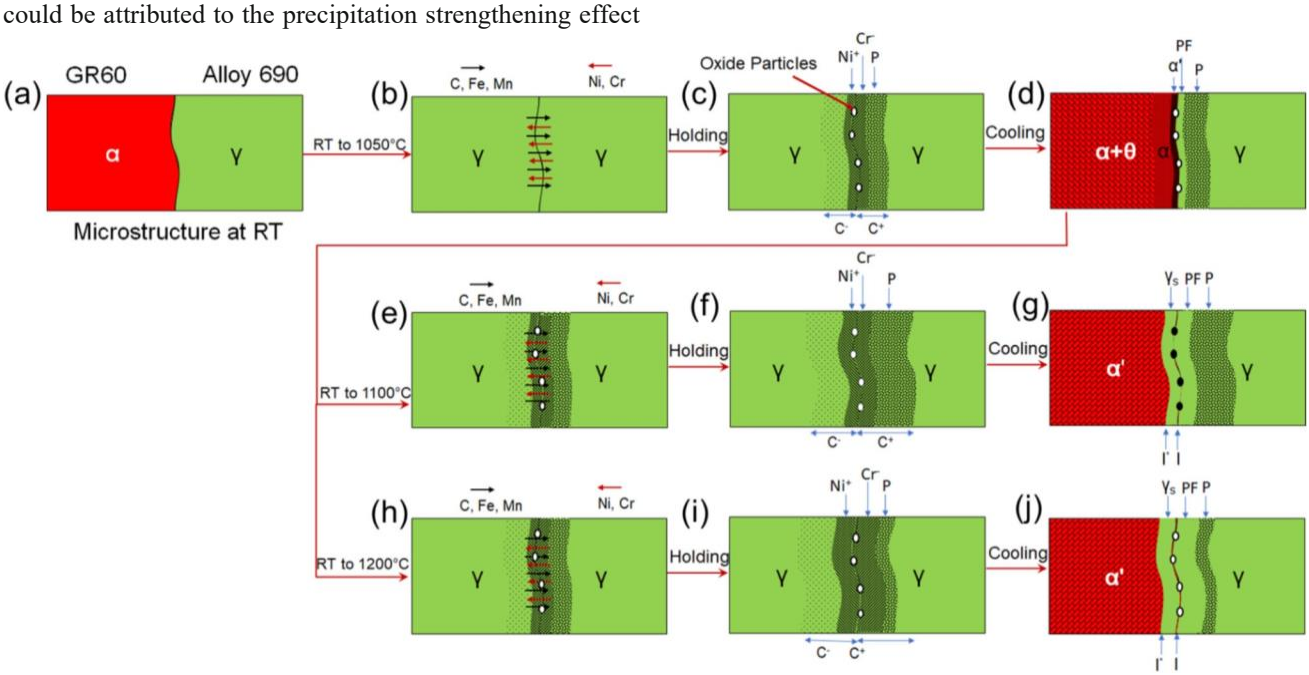

Fig. 13—Schematic depiction of microstructural evolution in Alloy 690/GR60 clad plate after different heat treatments. (P: Precipitation Zone; PF: Precipitate-free zone; c_s: stabilized FCC phase; a: Ferrite; h: Pearlite; a': Martensite, c: FCC-austenite; Ni^+ : Ni enriched zone; Cr: Cr depleted zone; C: decarburized zone, C^+: carburized zone, I: Actual Interface and I*: Apparent Interface.

The bulk regions of the GR60 layer in both solutionized specimens showed much higher hardness values as compared to the ASR specimen due to the presence of the martensitic microstructure. The hardness values in the GR60 layer of all three specimens were found to decrease toward the interface, which is mostly the direct manifestation of the decarburization of the GR60 layer due to the diffusion of carbon from GR60 toward Alloy 690. However, a slight increase in hardness values in the regions of the GR60 layer very close to the interface was observed in the solutionized specimens. On the other hand, the hardness values in ASR specimens tend to stabilize near the interface. The increase in hardness values of this region in solutionized specimens could be attributed to its significant enrichment of Cr and Ni due to diffusion from the Alloy 690 side, which could compensate the loss of solid solution strengthening due to depletion of carbon and ultimately resulted in the overall increase in hardness values. The width of this increased hardness region was larger in the case of the S1200 specimen as compared to the S1100 specimen, which closely correspond to the increased diffusion distance of solute atoms at (1473 K (1200 C)), in the S1200 specimen as shown in Figure 15. On the other hand, the loss of solid solution strengthening as a result of carbon depletion in ASR specimen could only be partially compensated due to the limited amount of diffusion of Cr and Ni from the Alloy 690 side. Therefore, the high hardness values in the ASR speci-

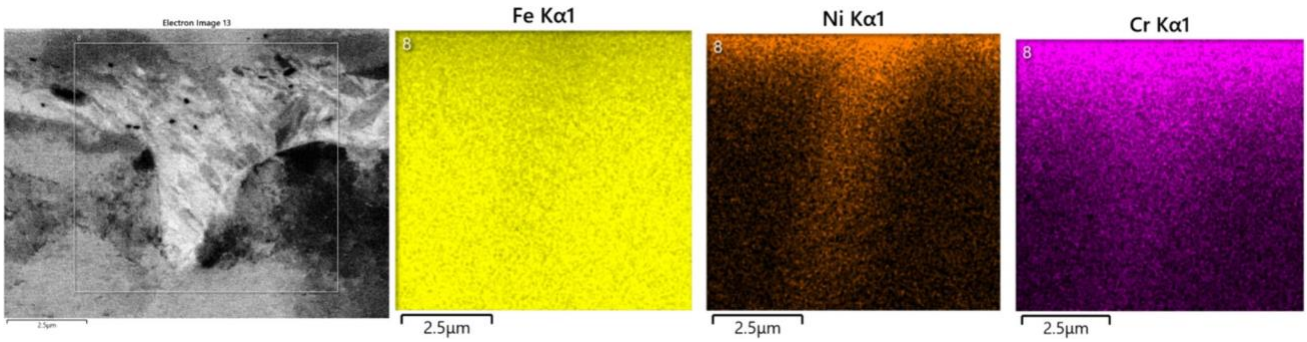

Fig. 14—EDS elemental maps within the martensite layer in as-rolled (ASR) clad specimen showing the grain boundary diffusion of Ni and Cr.

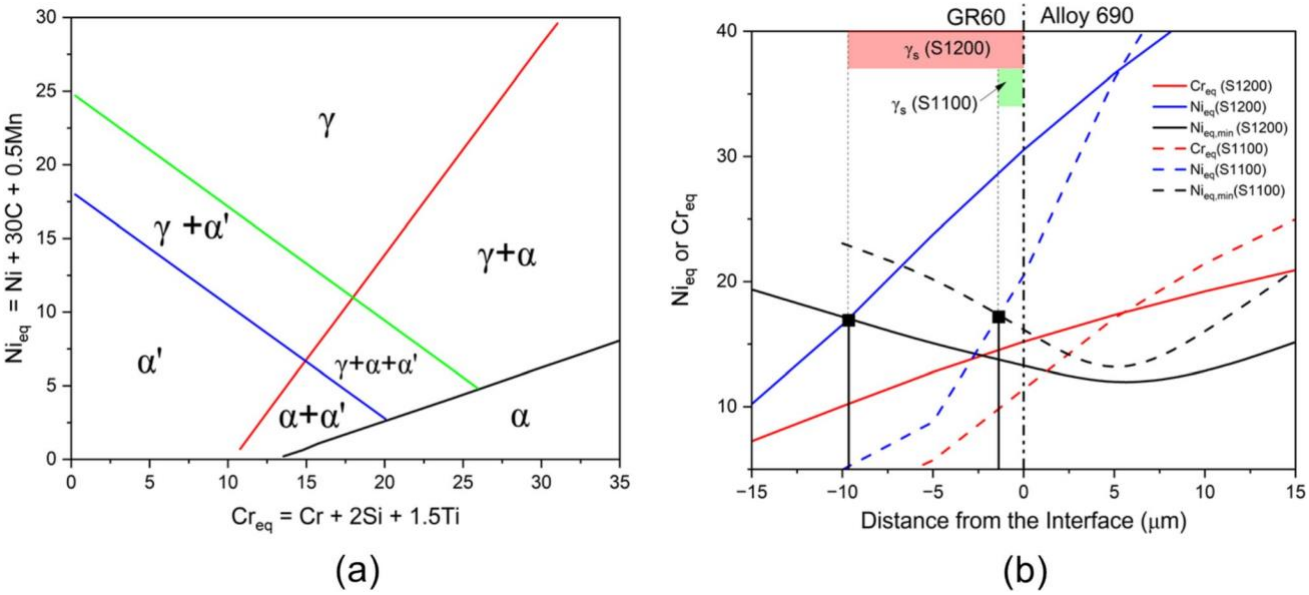

Fig. 15—(a) Showing stability of various phase in Fe-Cr–Ni alloy system at different Ni_{eq} and Cr_{eq} (adapted from Scheffler diagram Ref. [28]) and (b) curves showing variation of Cr_{eq}, Ni_{eq} across the interface and $Ni_{eq,min}$ required to stabilize the c-FCC phase at room temperature at the interface of hot roll bonded specimens solutionized at different temperatures.

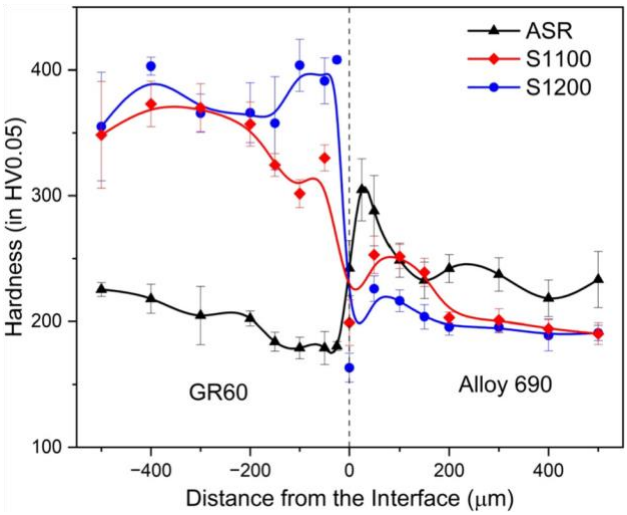

Fig. 16—Hardness variation across the Alloy 690/GR60 interface at different heat-treated conditions.

men tend to stabilize rather than show an increasing trend observed in the solutionized specimens.

The hardness value at the interface was highest in the case of the ASR specimen followed by S1100 specimen and lowest in the case of S1200 specimen. As observed in Figure 3, interface in ASR specimen consisted of a thin martensitic layer in GR60 side, which could have imparted a higher hardness at the interface in ASR as compared to the solutionized specimens. On the other hand, interface in both the solutionized specimens consisted of a precipitate-free c-FCC region, which showed a lower hardness value as compared to the thin martensitic layer present in the ASR specimen. Moreover, the grain coarsening at the precipitate-free region also gives rise to lower hardness values in the S1200 specimen as compared to the S1100 specimen.

In summary, the present study has demonstrated the suitability of VHRB technique for fabricating large size cost-effective Alloy 690 clad steel plate in industrial scale. The microstructural characterization has clearly revealed the possibility of tailoring the microstructure through post rolling heat treatments, which can be optimized through thermodynamic and long-range diffusion calculations. The effect of solutionization treatment on the mechanical properties of this clad plate is of utmost importance for utilizing this material for any industrial applications. This important aspect will be dealt in a separate manuscript at a later stage to complement this study. The effect of interfacial microstructure on the intergranular corrosion behavior of Alloy 690 clad plates also needs to be investigated before these clad plates can be successfully deployed for various industrial applications. Moreover, modern techniques such as cold spray deposition[30] for cladding of Alloy 690 over a carbon steel can also be explored. However, the challenges regarding complexity in powder preparation, sufficient thickness build up, porosity elimination in the deposited layer, sufficient interfacial strength, surface roughness and cost effectiveness of the overall process need to be compared with VHRB.

V. CONCLUSIONS

A triple-layered Alloy 690/GR60 steel clad plate was successfully fabricated through VHRB technique at 1323 K (1050 C) for the first time and the effect of solutionization heat treatments on the interfacial microstructure was studied. The major findings of the current study can be summarized as follows:

1. The interfacial microstructure of the as-rolled specimen revealed the presence of a precipitation zone containing $Cr_{23}C_6$ precipitates on the Alloy 690 side due to diffusion of C from GR60 layer.
2. Thermodynamic calculations showed possibility of minimization of these precipitates through solutionization treatment in the temperature range of 1373 K to 1573 K (1100 C to 1300 C).
3. Solutionization at 1373 K (1100 C) for 4 hours was found to increase the width of the $Cr_{23}C_6$ precipitation zone, although, further increase in solutionization temperature to 1473 K (1200 C) was observed to reduce the amount of $Cr_{23}C_6$ precipitates substantially.
4. Thermodynamic calculation revealed that at 1473 K (1200 C) temperature $Cr_{23}C_6$ is not an equilibrium phase and Cr_7C_3 phase is the only stable carbide. However, the experimental observation of a wider carbide precipitation-free (PF) region is essentially a manifestation of slower precipitation kinetics of Cr_7C_3 carbide phase in presence of Fe.
5. Diffusion of Ni and Cr from Alloy 690 to the GR60 side led to the formation of a thin martensitic layer at the interface in as-rolled clad plate due to increased hardenability. This distinct martensitic layer on GR60 side get stabilized into c_s phase due to Ni enrichment after solutionization treatments, which can be predicted well using Schaeffler diagrams.
6. The hardness value in Cr-carbide precipitation zone in all specimens is higher as compared to its adjacent PF region. Decarburization of GR60 layer near the interface resulted in dip in hardness near the interface, though, this effect gets less pronounced in solutionized specimens due to compensating solid solution strengthening by Cr and Ni enrichment from the Alloy 690. Bulk GR60 layer showed higher hardness after solutionization due to martensite phase formation while water quenching.

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CONFLICT OF INTEREST

On behalf of all authors, the corresponding author states that there is no conflict of interest.

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